

codehive

Generating and Modifying Persistent Data

1. Writing vs. Appending

In Python, the way you open a file determines what happens to existing data. Choosing the wrong mode can lead to data loss.

Append ('a'): High-safety. Adds new data to the end of the file without touching current content.

Write ('w'): Volatile. Wipes the file clean before adding new data.

2. The Append Mode ('a')

Use 'a' when you want to build logs or add entries to a list.

```
with open("log.txt", "a") as f:  
    f.write("New login detected\n")
```

If log.txt doesn't exist, Python will automatically create it for you.

3. The Overwrite Mode ('w')

Use 'w' when you want to replace an old file with a fresh version. **Warning:** This action cannot be undone.

```
with open("file.txt", "w") as f:
```

The moment this line executes, the content of file.txt is deleted (truncated to 0 bytes).

4. The Exclusive Create ('x')

Sometimes you want to ensure you don't overwrite a file by mistake. The 'x' mode is designed for this.

```
# Returns FileNotFoundError if "data.txt" exists  
f = open("data.txt", "x")
```

It is the safest way to initialize a **new** file project.

5. Writing Multiple Lines

The `write()` method doesn't add new lines automatically. You must include `\n` in your strings or use `writelines()`.

```
lines = ["First Line\n", "Second Line\n"]
with open("test.txt", "w") as f:
    f.writelines(lines)
```

6. Buffering and Flushing

Writing to a disk is slow, so Python collects data in a **buffer** before sending it. Data is usually written when you close the file.

If you need to force a write to the disk without closing the file, use `f.flush()`.

7. Encoding Considerations

When writing text, it is best practice to specify the encoding (usually UTF-8) to ensure your file is readable across all platforms.

```
with open("file.txt", "w", encoding="utf-8") as f:  
    f.write("Special characters: ñ, é")
```


8. Binary Writing ('wb')

To save images, PDF, or custom binary files, use 'wb'. You must provide **bytes** instead of strings.

```
binary_data = b'\x00\xff\x44'  
with open("output.bin", "wb") as f:  
    f.write(binary_data)
```

9. Exercise Check

Challenge

What happens to the original file content if you open it with 'w' mode?

☒ **The original content will be overwritten**

☐ New content is added after original content

10. Core Summary

- ✓ 'a' for safety (appending).
- ✓ 'w' for refreshing (overwriting).
- ✓ 'x' for security (error if exists).
- ✓ `write()` takes strings only.
- ✓ Always use `with` to ensure data saves.

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File Synthesis Series v1.1